MAT211: Calculus-II

Problem Sheet #02 **Topic: Limit & Continuity**

Date: February 27, 2026

Extra Practice (Not for marks)

1. Guess the limit then prove by $\delta - \epsilon$ method.

   (a) $\lim_{(x,y)\to(1,2)} \left(5x^3 - x^2y^2\right)$

   (b) $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}}$

   (c) $\lim_{(x,y)\to(2,1)} \frac{4-xy}{x^2+3y^2}$

   (d) $\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

   (e) $\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2}$

   (f) $\lim_{(x,y)\to(2,2)} \frac{x^3-y^3}{x^2-y^2}$

   (g) $\lim_{(x,y)\to(1,1)} \frac{2(x-y^2)}{x^2-y^4}$

   Hint: When I say "guess the limit," I mean you should first evaluate the expression by approaching the point along a simple path—such as changing only x or only y (i.e., along lines parallel to the axes). Since these problems do have limits, checking one such direction is enough to guess the correct value. Then you must verify that value using the ε–δ definition.

2. Consider the function:
$$f(x,y) = \frac{5xy^2}{x^2+y^2}.$$
   Show that the limit of $f(x,y)$ as $(x,y) \to (0,0)$ is 0.

3. Consider the function:
$$f(x,y) = \frac{xy^2}{x^2+y^4}.$$
   Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

4. Consider the function:
$$f(x,y) = \begin{cases} \frac{\cos y \sin x}{x}, & x \neq 0, \\ \cos y, & x = 0. \end{cases}$$
   Determine whether f is continuous at $(0,0)$. If not, can we redefine $f(0,0)$ to make f continuous at $(0,0)$? Prove using the definition of continuity.

1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

   Prove that f is not continuous at $(0,0)$. Consider the following sequences of points in $\mathbb{R}^2$.

2. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \dfrac{xy^2}{x^2+y^4}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

   Show that $\lim_{k\to\infty} f(x_k) = 0$ for any sequence $x_k \to (0,0)$ such that x_k (for $k \geq 1$) lies on a fixed line passing through $(0,0)$. Is f continuous at $(0,0)$?

3. Let

$$g(x,y) = \begin{cases} x \sin\left(\dfrac{1}{y}\right), & y \neq 0, \\ 0, & y = 0. \end{cases}$$

   Determine the points $(x,y) \in \mathbb{R}^2$ where g is continuous.

Theorem

1. Suppose f and g map a neighborhood of $a \in \mathbb{R}^n$ (with the possible exception of the point a itself) to $\mathbb{R}^m$ and k maps the same neighborhood to $\mathbb{R}$. Suppose

$$\lim_{x \to a} f(x) = \ell, \qquad \lim_{x \to a} g(x) = m, \qquad \text{and} \qquad \lim_{x \to a} k(x) = c.$$

Then

$$\lim_{x \to a} \big(f(x) + g(x)\big) = \ell + m,$$
$$\lim_{x \to a} \big(f(x) \cdot g(x)\big) = \ell \cdot m,$$
$$\lim_{x \to a} \big(k(x)f(x)\big) = c\ell.$$

Solution: Theorem 3.2 from *Multivariable Mathematics Linear Algebra, Multivariable Calculus, and Manifolds* (Theodore Shifrin).

2. Suppose $U \subset \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^m$. Then f is continuous at a if and only if for every sequence $\{x_k\}$ of points in U converging to a, the sequence $\{f(x_k)\}$ converges to $f(a)$.

Solution: Proposition 3.6 from Shifrin.

3. Let $U \subset \mathbb{R}^n$ be an open set. The function $f : U \to \mathbb{R}^m$ is continuous if and only if for every open subset $V \subset \mathbb{R}^m$, $f^{-1}(V)$ is an open set (i.e., the preimage of every open set is open).

Solution: Proposition 3.4 from Shifrin.

4. Suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous. Then for any $c \in \mathbb{R}^m$, the level set

$$f^{-1}(\{c\}) = \{x \in \mathbb{R}^n : f(x) = c\}$$

is a closed set.

Solution: Corollary 3.7 from Shifrin.

Iterated Limit

Suppose that

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

and the one-dimensional limits

$$\lim_{x\to a} f(x,y) \quad \text{and} \quad \lim_{y\to b} f(x,y)$$

both exist. Then

$$\lim_{x\to a}\left[\lim_{y\to b} f(x,y)\right] = \lim_{y\to b}\left[\lim_{x\to a} f(x,y)\right] = L.$$

(a) Prove the above theorem.

(b) Provide an example of a function $f(x,y)$ such that

$$\lim_{x\to 0}\left[\lim_{y\to 0} f(x,y)\right] = 1 \quad \text{but} \quad \lim_{y\to 0}\left[\lim_{x\to 0} f(x,y)\right] = -1.$$

(c) Give an example of a function $f(x,y)$ such that

$$\lim_{x\to 0}\left[\lim_{y\to 0} f(x,y)\right] = \lim_{y\to 0}\left[\lim_{x\to 0} f(x,y)\right] = 0$$

but $f(x,y)$ does not tend to a limit as $(x,y)\to(0,0)$.

(d) Give a function $f(x,y)$ such that $f(x,y)\to 0$ as $(x,y)\to(0,0)$, but

$$\lim_{y\to 0}\left[\lim_{x\to 0} f(x,y)\right] \neq \lim_{x\to 0}\left[\lim_{y\to 0} f(x,y)\right].$$